Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework

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Outline

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Motivation

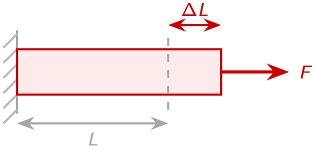
- The cubic stiffness constants C_{11}, C_{12}, and the derived Young's modulus E and Poisson's ratio ν, are the main design criteria for structural alloys.
- Measuring them is slow and expensive. Classical molecular dynamics recovers the same properties at negligible cost, but only if an accurate interatomic potential is available.
- For metals that potential is usually the Modified Embedded Atom Method (MEAM) [1, 2], the standard semi-empirical form.

Objective

Build MEAM potentials automatically for a chosen set of elements, with no manual parameter fitting and no experimental input.

A material's elastic response is captured by two numbers. For a cubic crystal both follow from the stiffness constants C_{11} and C_{12}, the quantities this work predicts: C_{11} is the stress needed to stretch the crystal along one axis; C_{12} is the sideways stress that builds up while it stretches.

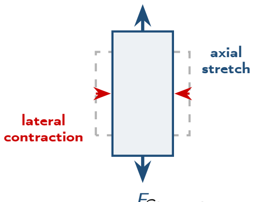
E: resistance to stretching



$$E = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta L/L}$$

Large E means stiff: steel ≈ 200, Al $\approx 70\,\mathrm{GPa}$.

ν: sideways thinning when pulled



$$\nu = -\frac{\varepsilon_{\text{lateral}}}{\varepsilon_{\text{axial}}}$$

Stretch it and it thins: most metals $\nu \approx 0.3$.

$$E = \frac{(C_{11}-C_{12})(C_{11}+2C_{12})}{C_{11}+C_{12}}, \quad \nu = \frac{C_{12}}{C_{11}+C_{12}};$$ stability requires $C_{11} > C_{12}$ (Born–Cauchy).

Predicting C_{11}, C_{12} from Atoms: Why We Need Potentials and DFT

Interatomic potential. Closed-form approximation of the many-body potential energy $U(\{\mathrm{r}_i\})$; forces obtained as $\mathrm{F}_i = -\nabla_i U$.

Historical development:

- Pairwise (Lennard-Jones, Morse): too crude for metals.
- Embedded Atom Method (1984): good for FCC metals.
- **Modified EAM** (MEAM) [1]: adds angular terms and covers BCC, HCP and mixed bonding.

Density-functional theory (DFT). First-principles electronic-structure method (Kohn–Sham equations [3]); plane-wave + pseudopotential basis in Quantum Espresso [4].

Quantities computed:

- Cohesive energy E_{coh}
- Equilibrium lattice a_0, bulk modulus B
- Binary formation energies $E_{\mathrm{form}}(i,j)$
- Single-crystal elastic constants C_{11}, C_{12}

DFT: high accuracy, limited scale ($\leq 10^3$ atoms).
MEAM/MD: empirical, large scale ($\geq 10^6$ atoms).
DFT provides the reference dataset against which the MEAM potential is calibrated.

Why No Existing Potential Suffices: The MEAM Coverage Gap

Published MEAM libraries cover disjoint subsets of elements; no single file spans the 12 elements treated in this work.

Library	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn [5]	✓					✓						✓
AlSiCuMgFe [6]	✓			✓	✓	✓				✓		
CuMo [7]				✓				✓				
FeMnNiTiCuCrCoAl [8]	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Limitation of direct merging: cross-interaction parameters between elements drawn from distinct libraries are not defined; manual refitting does not scale to $\binom{12}{2} = 66$ binary pairs.

Pure machine-learning potentials [9–11]: require $10^4 - 10^5$ DFT configurations and forfeit the physical interpretability of MEAM.

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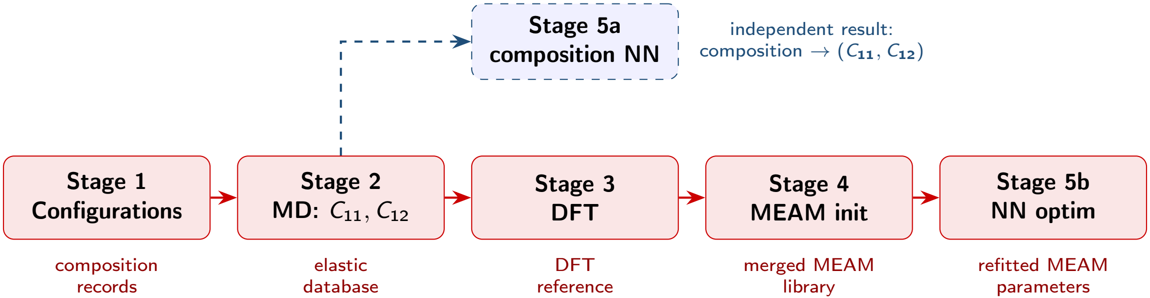
Method: the five-stage pipeline

Proposed Methodology: DFT-Informed NN Optimisation



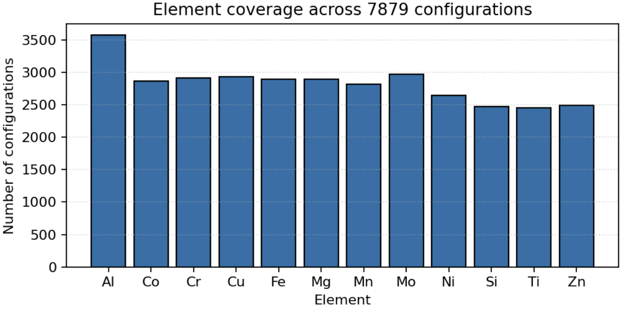
The four published libraries are merged into a single 12-element file; cross-terms are refitted by inverting an NN surrogate of the MEAM $\to (C_{11}, C_{12})$ map by gradient descent, with single-element parameters anchored to DFT references.

MD = Molecular Dynamics; DFT = Density-Functional Theory; NN = Neural Network.

Stage 1: Alloy Configuration Generation

Sampling.
Uniform draws on the 12-element composition space (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn);
Al-weighted to oversample the aluminium-alloy region.

Dataset statistics.
7879 configurations generated;
7815 retained after the physicality filter ($0 < E < 400\,\mathrm{GPa}$, $0 < \nu < 0.5$).



Element-occurrence distribution over the full generated dataset.

Stage 2: Molecular-Dynamics Elastic Database

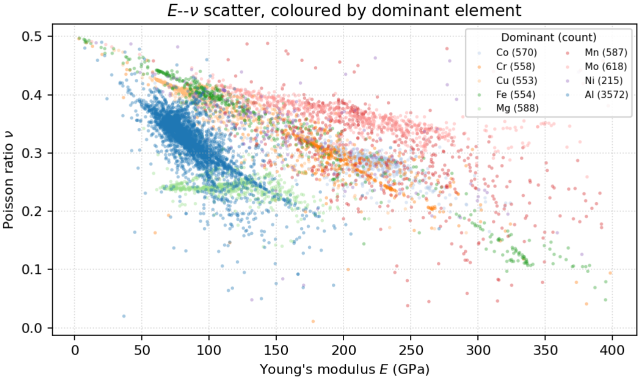
Implementation. [12]
LAMMPS molecular dynamics with the 4 published MEAM potentials.

Per-configuration protocol:

- $\sim$4 nm cubic supercell
- Anisotropic energy minimisation
- Axial strain $\delta = 10^{-3}$ applied along $\pm x, \pm y, \pm z$
- Central-difference $\rightarrow C_{11}, C_{12}$

	Median	Std.	Range
E (GPa)	103.6	69.1	3 – 398
ν	0.326	0.063	0.01 – 0.50

$N = 7815$ retained configurations.



E–ν distribution, coloured by dominant element.

Stage 3–4: DFT References → MEAM Initialisation

DFT references + existing MEAM-file cross-interactions → **MEAM initialisation** α + mixing sign → **12-element MEAM library**

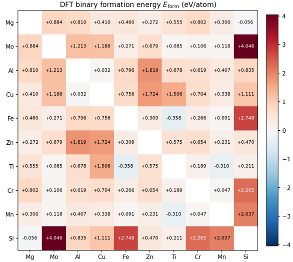
Single-element terms

Each element's $(a_0, E_{\mathrm{coh}}, B)$ pins the universal binding curve (spacing, well depth, curvature) and fixes the MEAM decay exponent

$$\alpha = \sqrt{9B\Omega/E_{\mathrm{coh}}} \qquad (\Omega : \text{atomic volume}).$$

Scope. 12 elements (FCC/BCC/HCP/diamond); $\alpha \approx 4.5$ (Al, Si) to 10.9 (Zn). PBE–GGA plane waves (Quantum Espresso); a_0, B from an equation-of-state fit.

Cross-terms



Binary formation energies $E_{\mathrm{form}}(i, j)$ (eV/atom): negative = exothermic mixing, positive = limited solubility; the sign seeds each cross-term.

03

Results

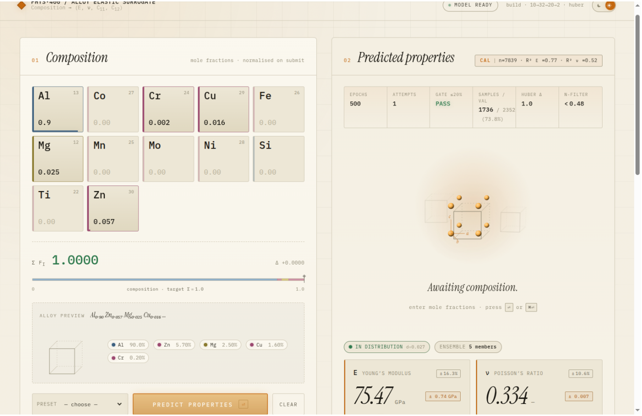
TensorFlow / Keras

Architecture. $12 \to 32 \to 20 \to 2$ fully-connected MLP with ReLU activation; Huber loss on $\hat{C}_{11}, \hat{C}_{12}$; deep ensemble of five seeds [13] for epistemic uncertainty.

Result

Validation Set (30 % of the dataset): $R^2_E = 0.77,\ R^2_\nu = 0.52$.

Al 7075 Example: $\hat{E} = 75.5\,\text{GPa}$, $\hat{\nu} = 0.334$ (published: 71–72 GPa, $\nu \approx 0.33$; within $\sim$5 %).



Interactive composition predictor.

Stage 5b: NN-Driven MEAM Parameter Optimisation

TensorFlow + LAMMPS

Closed-loop procedure

1. **Surrogate.** A small MLP learns (MEAM params) $\to (C_{11}, C_{12})$.
2. **Fit.** Train it with Adam [14] on a Huber loss.
3. **Invert.** Descend through the *frozen* surrogate to the parameters that hit the target (C_{11}, C_{12}).
4. **Loop.** Re-evaluate in LAMMPS, append samples, retrain.

Adam adapts the step size per parameter; the Huber loss keeps outlier (Mg/Zn) configurations from dominating the fit.

Surrogate Huber loss decays 0.30 $\to$ 0.16 over training.

Result: a working proof of concept

Tested on **30 unseen alloys**: **11** were physically stable, and **4** matched the MD reference within $\sim$20 %. Accuracy grows with more data and longer training.

External Validation: Quantitative Summary

By material family ($N = 42$)

Family	N	E MAPE (%) ML	E MAPE (%) MEAM	ν MAPE (%) ML	ν MAPE (%) MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Overall	42	**8.5**	42.2	**7.2**	7.1

Aluminium by AA series ($N = 33$)

Al series	N	$\langle E_{\mathrm{lit}} \rangle$ (GPa)	E MAPE (%) ML	E MAPE (%) MEAM
2xxx (Al–Cu)	11	72.7	1.3	8.0
5xxx (Al–Mg)	9	70.5	2.1	7.9
6xxx (Al–Mg–Si)	7	68.9	3.8	4.2
7xxx (Al–Zn–Mg)	6	71.8	2.8	9.3
All aluminium	33	71.1	**2.3**	7.4

Both surrogates reproduce aluminium Young's modulus to within a few percent (ML 2.3 %, MEAM 7.4 %) across all four alloy series; error grows only *off* the Al-rich design domain, steeply for MEAM and gently for the composition NN.

42 basis-covered alloys; MAPE vs experimental data from ASM/MatWeb [15]. Unphysical MEAM runs excluded.

04

Discussion & summary

Summary & Further Studies

Achieved

- ✓ **Automated 12-element MEAM pipeline.** An end-to-end workflow that assembles a multi-element potential.
- ✓ **Aluminium-alloy elastic prediction.** E, ν and C_{11}, C_{12} obtained from composition.
- ✓ **DFT-seeded, NN-trained MEAM.** First-principles references refit through a neural surrogate.

What comes next

1. **More data coverage.** Wider sampling across the composition space and crystal structures.
2. **More NN training epochs.** Deeper surrogate convergence before each refit.
3. **Denser atomic-cell DFT.** Finer cells and supercell sampling for sharper references.
4. **Inverse design.** Specify a target (E, ν) and invert the composition surrogate to propose candidate alloys.

The pipeline already runs end to end; what is left is more data, longer training, denser DFT references, and running the predictor in reverse.

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